

Chapter 16

Long-Run Growth

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- ➊ A Brief History of Economic Growth
- ➋ Sources of Economic Growth
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Terminologies

- Output growth: The growth rate of the output of the *entire* economy

$$g_t = \frac{Y_t - Y_{t-1}}{Y_{t-1}}.$$

- Per-capita output growth: The growth rate of output per *person* in the economy

$$y_t = \frac{Y_t}{N_t}.$$

$$\tilde{g}_t = g_t - g_{N,t}.$$

- Labor productivity growth: The growth rate of output per worker

$$\hat{y}_t = \frac{Y_t}{L_t}.$$

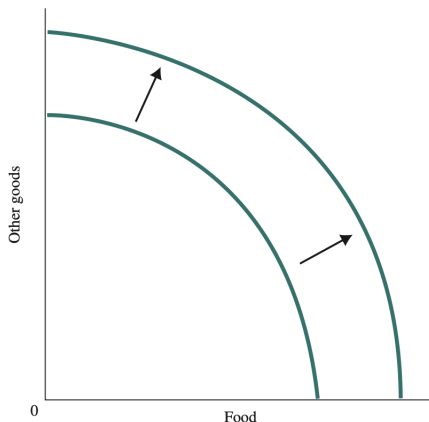
$$\hat{g}_t = g_t - g_{L,t}$$

From Agriculture to Industry

- Recall that economic growth is quite a recent phenomenon from a broader historical perspective throughout the past 2,000 years or so.
- Before the Industrial Revolution in Great Britain, every society in the world was agrarian.
- Beginning in England around 1750, technical change and capital accumulation increased productivity significantly in two important industries: agriculture and textiles.
 - New and more efficient methods of farming were developed.
 - New inventions and new machinery in spinning and weaving meant that more could be produced with fewer resources.
- Growth brought it with new products, more output, and wider choice.

Economic Growth: Expansion of PPF

- The case of the productivity growth such as Industrial Revolution can be represented graphically.



Aggregate Production Function

- As labor and capital increase, so will output.
- Mathematical relationship stating that total GDP Y depends on the total amount of labor used L and the total amount of capital K used.
- Example: Cobb-Douglas Production Function

$$Y = 3 \times K^{1/3} L^{2/3}.$$

- The following factors are the sources of economic growth
 - Increase in the quantity of the labor supply
 - Increase in the quantity of the physical capital
 - Increase in the quality of the labor supply: human capital (education and health)
 - Increase in the quality of capital (embodied technical progress)
 - Disembodied technical progress

Technical Progress

- Embodied technical change: technical change that results in an improvement in the quality of capital.
 - Example: faster computer chips, which are then incorporated into computers.
- Disembodied technical change: technical change that results in a change in the *production process* (system).
 - In some situations we can achieve higher levels of output over time even if the quantity and quality of labor and capital don't change.
 - Example: how to better organize the plant floor, how to manage the labor force, improvements in information and accounting systems, incentive systems, etc.

Pollution and Economic Growth

- The relationship between smoke concentration and per-capita GDP is an inverted U.
 - As countries grow wealthier, smoke increases and then declines.

